ENGL220 | SPRING 2026 | DR. NAQAA ABBAS

WALKABILITY IN EDUCATION CITY

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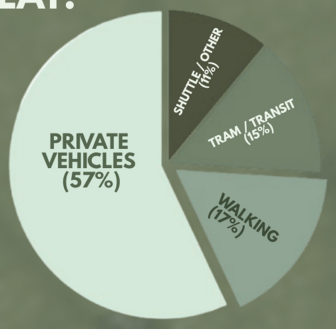
THE PROBLEM

WALKING IN THE HEAT:

Despite a 24-station tram, 40˚C+ [5] temperatures and unshaded walkways make walking across Education City's 12 km² campus unbearable, forcing most to drive



Transportation mode distribution among the Education City population [1]

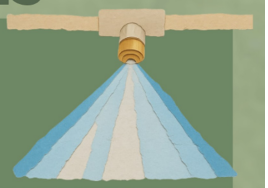
QNV 2030: MAJOR CRITERIA

- Environmental Sustainability : NDS3 25% emissions reduction target[3]
- Sustainable Transport: Encourage walking over driving[3]
- Human Development: Physical activity and community well-being[3]
- Urban Planning: Achieve 60~70% pedestian shade[2]

SOLUTIONS

SOLAR-POWERED MISTING SYSTEMS

High-pressure misting nozzles powered by solar panels that lower surrounding temperatures



PARK AND PLANT

Plant tree native species such as Sidra and Ghaf trees along walkways to form continuous shaded routes

COOL PAVEMENT COATINGS ON WALKWAYS

Reflective coatings that reduce the amount of heat absorbed by the pavements

CLIMATE-CONTROLLED WALKWAYS

Enclosed, air-conditioned walkways that provide stable internal temperature

CHOSEN SOLUTION: PARK AND PLANT

AN EXISTING QF COMMUNITY-LED PROGRAM [4]

What: Planting native Sidra and Ghaf trees along walkways.

Why: To create continuous, shaded corridors.

How: Cooling the air through sun blockage.

Our recommendation: Increase the number of trees planted along walkways

ADVANTAGES

- CLIMATE-PROOF: EFFECTIVE COOLING IN ANY HUMIDITY.
- ECO-FRIENDLY: SEQUESTERS CARBON AND CLEANS THE AIR.
- FEASIBLE: LEVERAGES ACTIVE QF STAFFING AND RESOURCES.
- SUSTAINABLE: LOW-WATER, NATIVE SPECIES ADAPTED TO QATAR.
- COMPOUNDING: SHADE AND IMPACT GROW YEAR AFTER YEAR.

CRITERIONS CONSIDERED

- THERMAL COMFORT
- SUSTAINABILITY
- COVERAGE & SCALABILITY
- FEASIBILITY

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REFERENCES